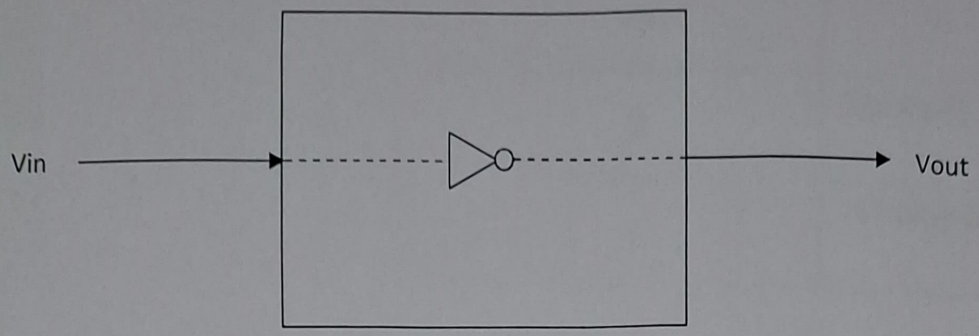
Class	:	BE 7
Batch	:	S7
Roll. No	:	42365
ABC ID	:	553725367025
Assignment No.	:	B.1.a
Assignment Name	:	CMOS INVERTER & Dynamic Power Dissipation Analyses
Date Of Performance	:	

SYMBOL:-



$Vout = \overline{Vin}$

MOSFET LEVEL SCHEMATIC & THEORY

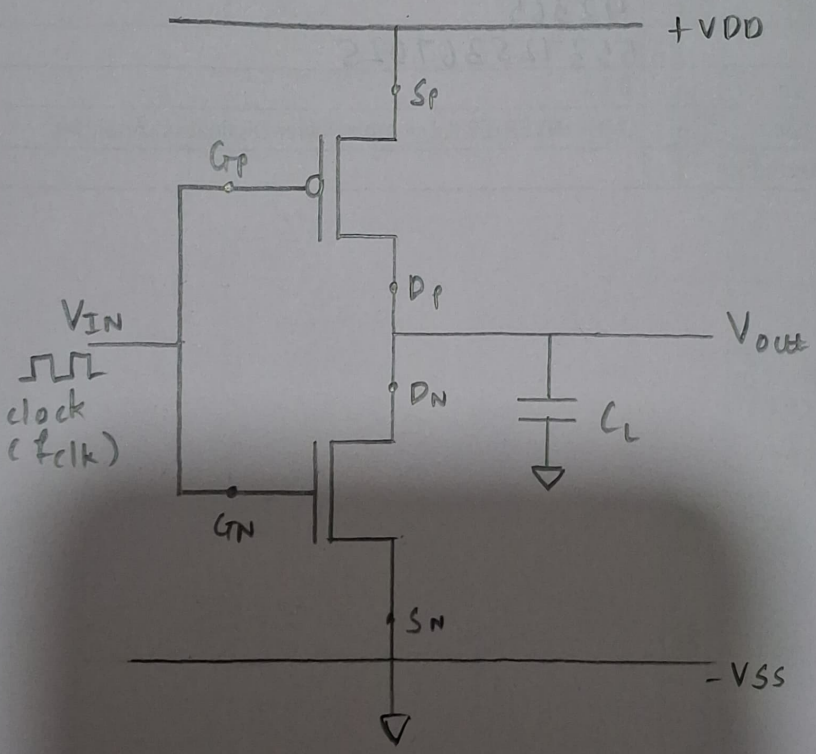


Fig: MOSFET Level Schematic of CMOS Inverter

DYNAMIC POWER DISSIPATION ANALYSES

(I) Effect of change in C_L : f_{clk} = 5 GHz , V_{DD} = 1.2 V (90 nm Foundry)

SR.NO.	C_L (pF)	P_{dyn} (µW)
1)	0.001	13.088 µW
2)	0.002	19.238 µW
3)	0.0005	10.066 µW

(II) Effect of change in f_{clk} : C_L = 1 fF , V_{DD} = 1.2 V (90 nm Foundry)

SR.NO.	f_{clk} (GHz)	P_{dyn} (µW)
1)	5	12.733 µW
2)	2.5	6.846 µW
3)	10	24.318 µW

DYNAMIC POWER DISSIPATION ANALYSES

V_{DD} (V)	P_{dyn} (μW)
1.2 (90 nm)	12.733 μW
2 (180 nm)	42.835 μW

Conclusions:-

1) Drawn the LAYOUT for CMOS Inverter using 90 nm & 180 nm Foundry.
2) Simulated LAYOUT to observe w/f's & verified functionality.
3) Being a **Pure-CMOS System** (PMOS // NMOS & CMOS INVERTER) , it gives both **S-1** & **S-0** as O/P.
4) Appreciated the validity of the mathematical model

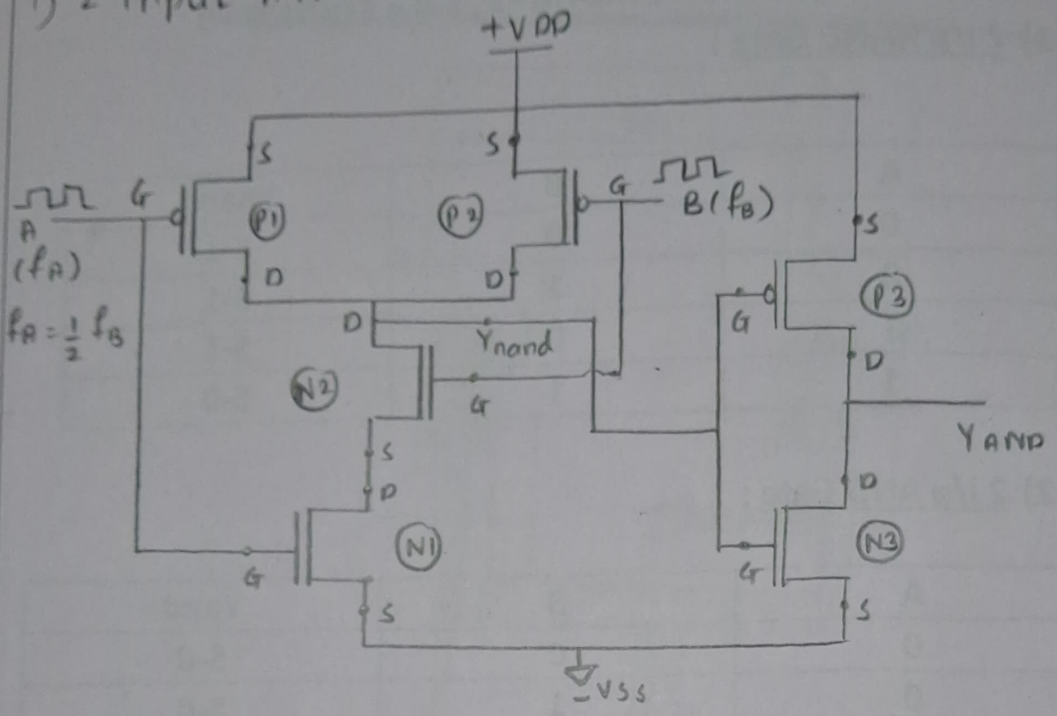
$$P_{dynamic} = C_L * (Vdd)^2 * f_{clk}$$

5) Found a reduction in $P_{dynamic}$ by using a better Foundry i.e., **90 nm** instead of **180 nm**
6) Learnt that the presence of spikes in O / P waveform at Switching instants indicate the inability of the MOSFETs to switch at GHz frequencies.

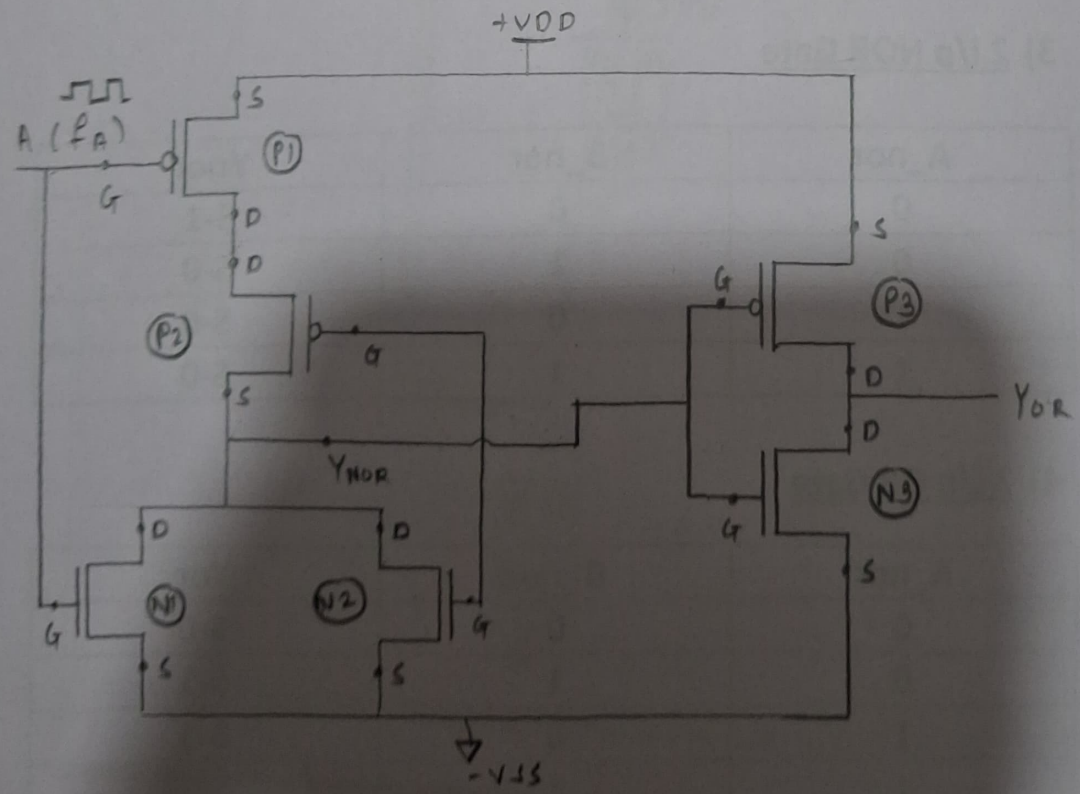
Class	:	BE 7
Batch	:	S7
Roll. No	:	42365
ABC ID	:	553725367025
Assignment No.	:	B.1.b , B.1.c
Assignment Name	:	CMOS 2 i/p NAND-AND , CMOS NOR-OR Logic Gates (90nm Foundry)
Date Of Performance	:	

MOSFET LEVEL SCHEMATIC & THEORY

1) 2 input AND Gate

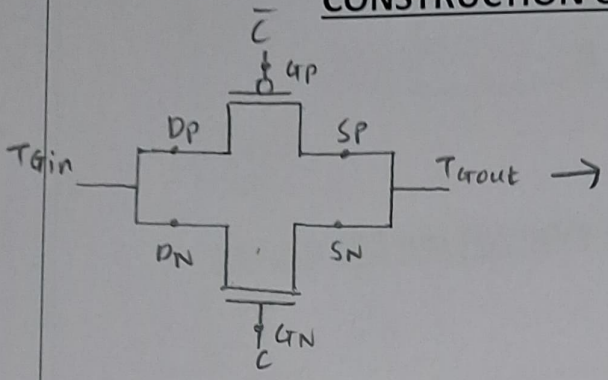


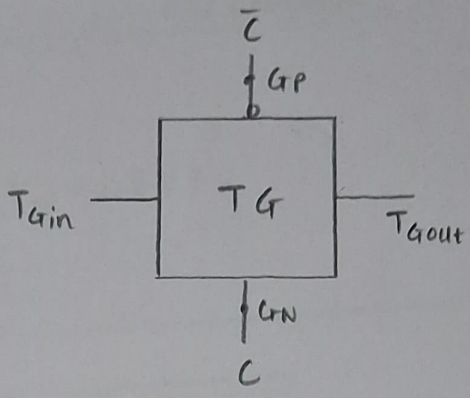
2) 2 input OR Gate

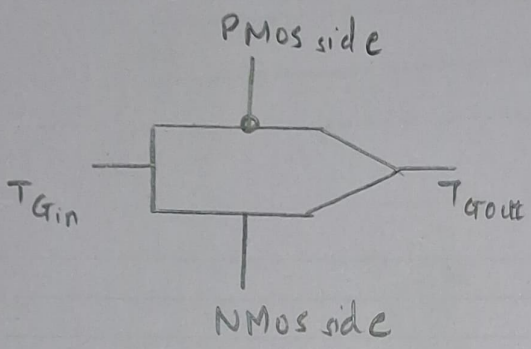


Class	:	BE7
Batch	:	S7
Roll. No	:	42365
ABC ID	:	553725367025
Assignment No.	:	B.1.d
Assignment Name	:	CMOS TRANSMISSION-GATE (TG)
Date Of Performance	:	

CONSTRUCTION & SYMBOLS OF TG







operation:

C = 0 (NMOS side)

$\bar{C}$ = 1 (PMOS side)

TG = OFF

C = 1 (NMOS side)

$\bar{C}$ = 0 (PMOS side)

TG = ON

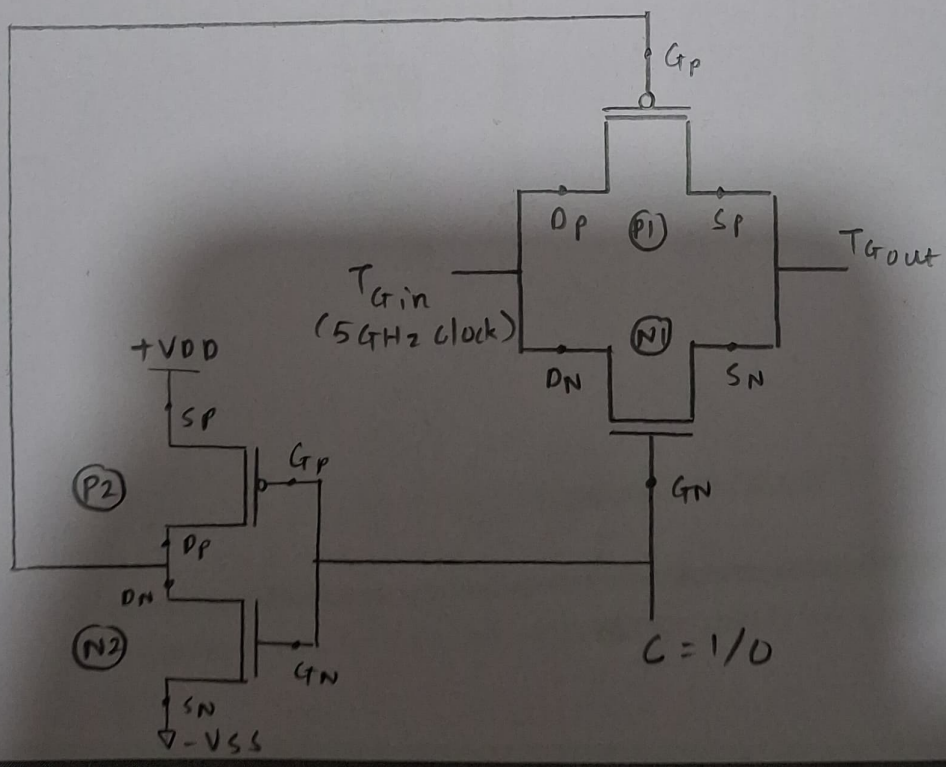
C = Control Signal (Applied to Gate of NMOS)

$\bar{C}$ = Inverted Control Signal (Applied to Gate of PMOS)

C= 1 : NMOS = PMOS = ON

O/P Node is Connected to I/P Node

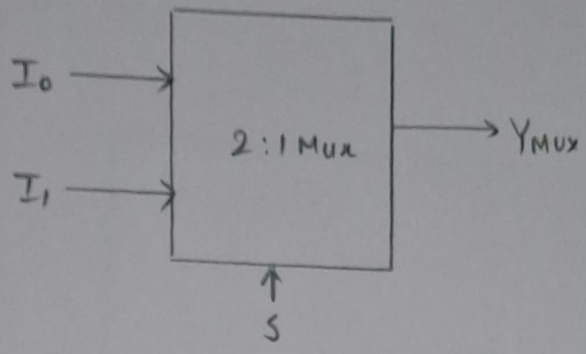
MOSFET-LEVEL SCHEMATIC of TG



Class	:	BE7
Batch	:	S7
Roll. No	:	42365
ABC ID	:	553725367025
Assignment No.	:	B.2
Assignment Name	:	2:1 MULTIPLEXER Using TRANSMISSION GATE (TG)
Date Of Performance	:	

SYMBOL , TRUTH-TABLE , SYMBOL-LEVEL SCHEMATIC

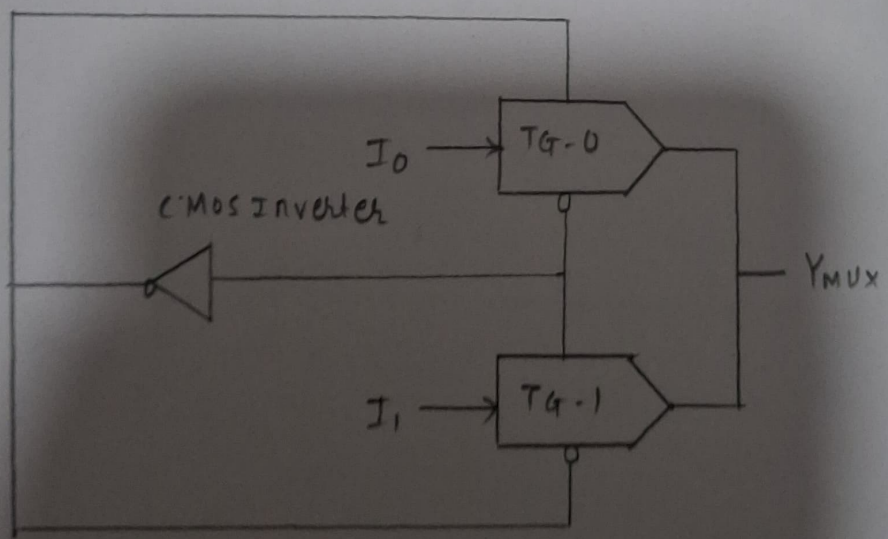
1) 2 : 1 MUX



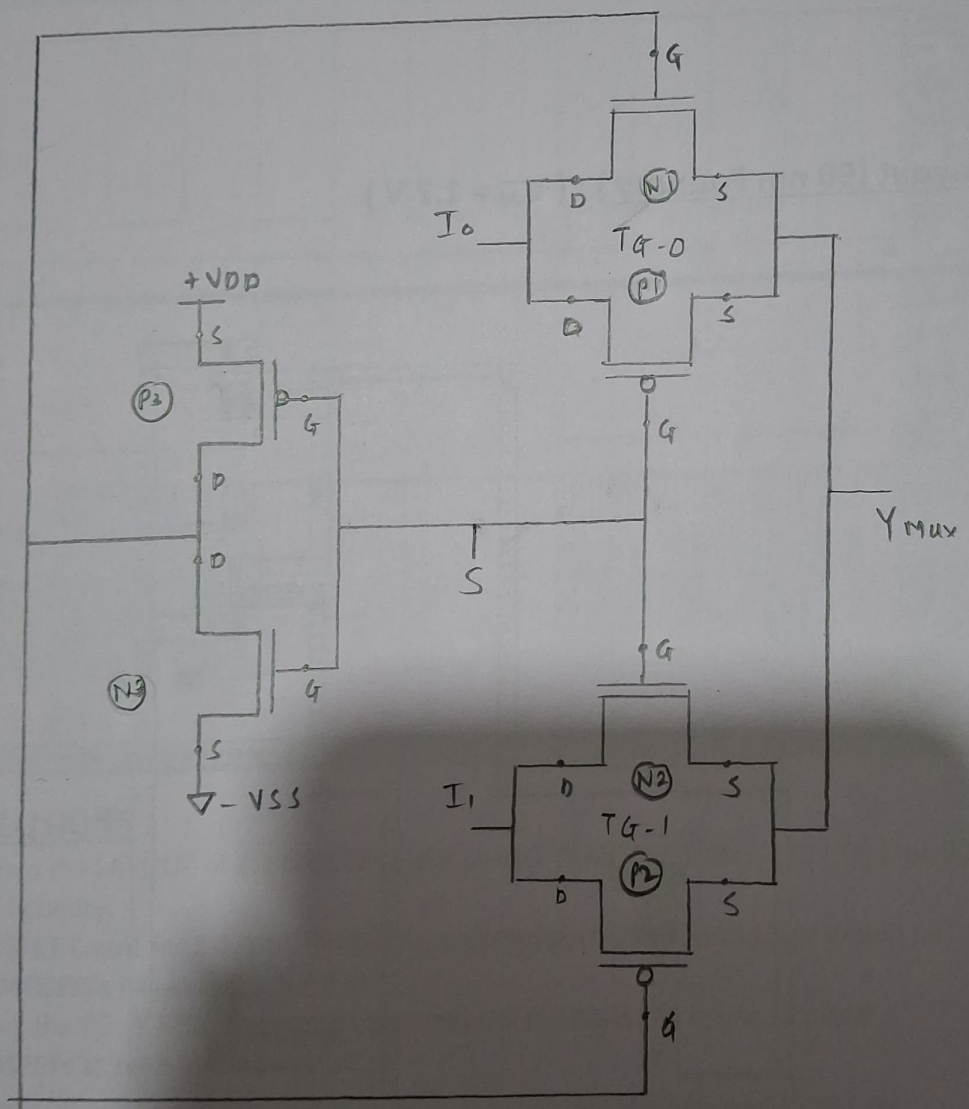
2) Truth table

S	Y_{MUX}
0	I_0
1	I_1

3) symbol level schematic



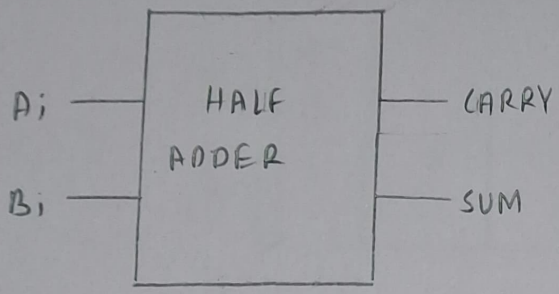
MOSFET-LEVEL SCHEMATIC of 2:1 MUX Using TG



Class	:	BE7
Batch	:	S7
Roll. No	:	42365
ABC ID	:	553725367025
Assignment No.	:	B.3.a
Assignment Name	:	CMOS HALF-ADDER (Using MUX & MUX – Using TG)
Date Of Performance	:	

BLOCK DIAGRAM , TRUTH – TABLE OF HALF – ADDER

Block Diagram

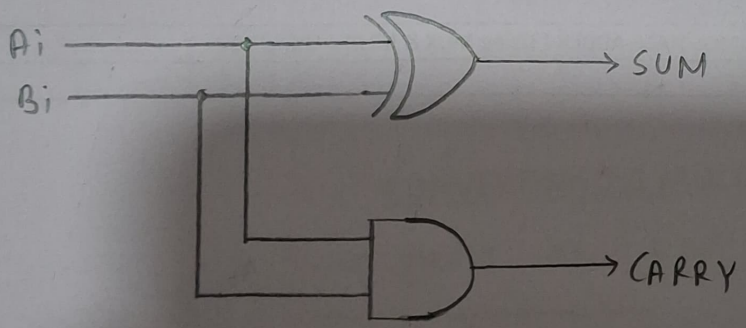


Truth table

Ai	Bi	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

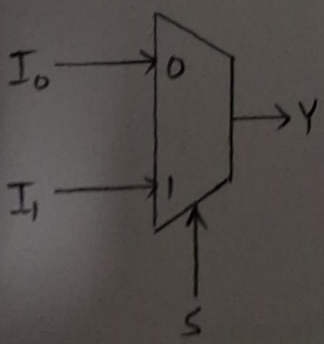
$SUM = A_i \oplus B_i$ $CARRY = A_i \cdot B_i$

GATE – LEVEL SCHEMATIC OF HALF – ADDER



2:1 MULTIPLEXER SYMBOL & TRUTH-TABLE

symbol



Truth table

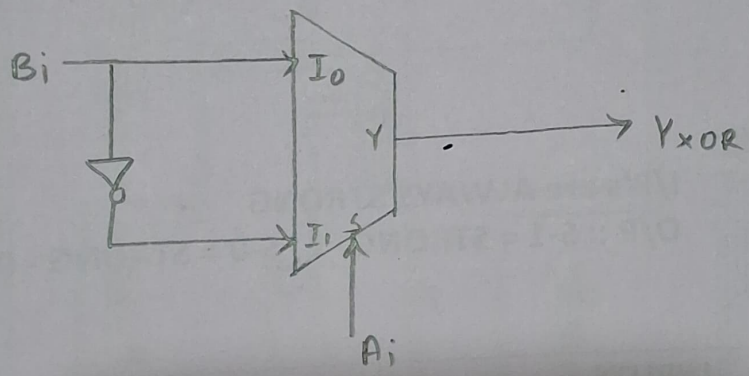
S	Y
0	I_0
1	I_1

CONVERTING 2:1 MULTIPLEXER INTO EX-OR GATE

Truth table of EXOR Gate

	Ai	Bi	Y_{XOR}	
Ai=0	0	0	0	$Y_{XOR} = Bi$
	0	1	1	
Ai=1	1	0	1	$Y_{XOR} = \overline{Bi}$
	1	1	0	

∴ Give Ai as select line
Bi as I_0 i/p of mux
$\overline{Bi}$ as I_1 i/p of MUX

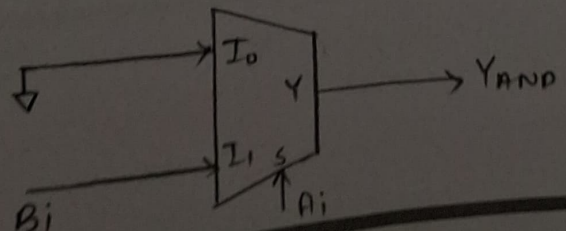


CONVERTING 2:1 MULTIPLEXER INTO AND GATE

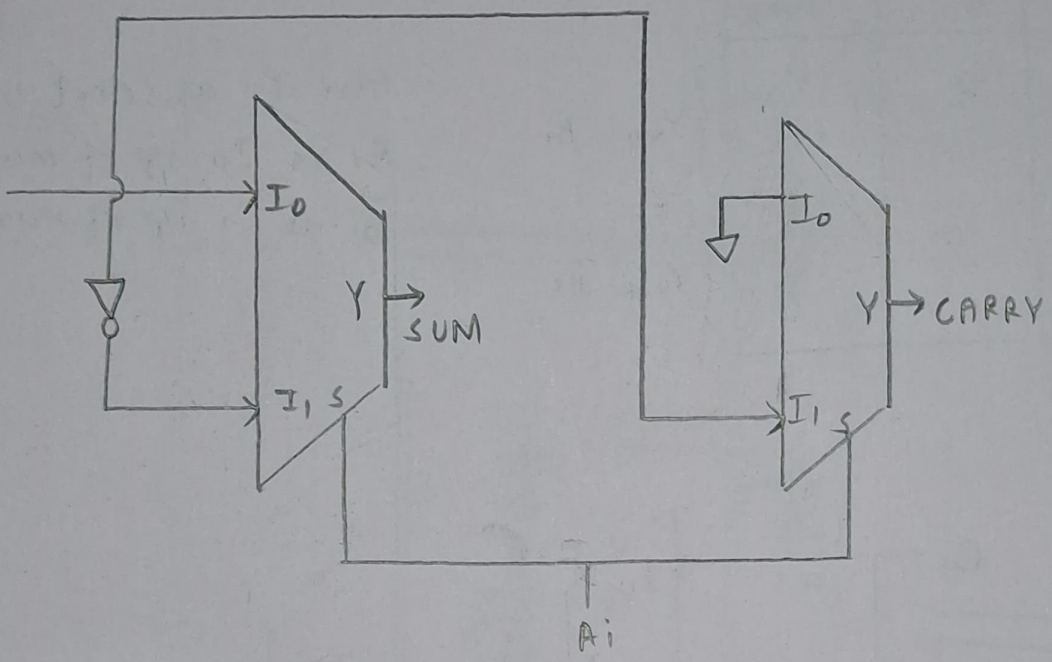
Truth table of AND gate

	Ai	Bi	Y_{AND}	
Ai=0	0	0	0	$Y_{AND} = 0$
	0	1	0	
Ai=1	1	0	0	$Y_{AND} = Bi$
	1	1	1	

∴ Give Ai as select line
0 as I_0 i/p of MUX
Bi as I_1 i/p of MUX



MULTIPLEXER – LEVEL SCHEMATIC OF HALF - ADDER



Truth Table:-

I/P's are ALWAYS STRONG

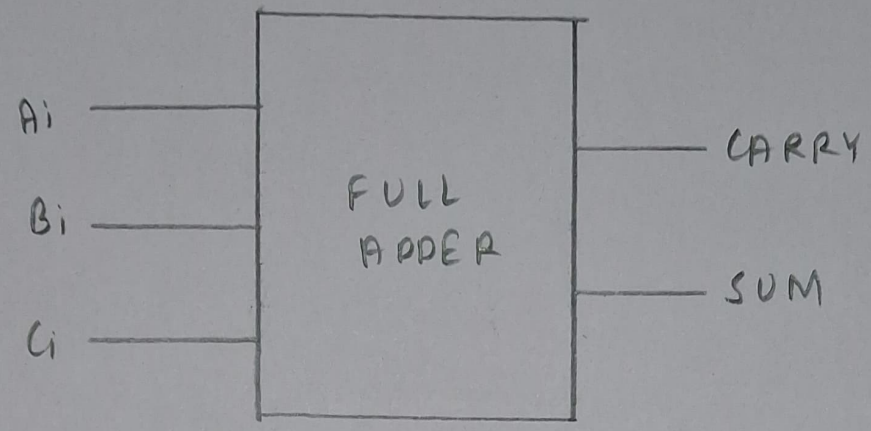
O/P :: S-1 = STRONG -1 , S-0 = STRONG - 0

INPUTS		OUTPUTS	
Ai	Bi	SUM	CARRY
0	0	S-0	S-0
0	1	S-1	S-0
1	0	S-1	S-0
1	1	S-0	S-1

Class	:	BE7
Batch	:	S7
Roll. No	:	42365
ABC ID	:	553725367025
Assignment No.	:	B.3.b
Assignment Name	:	CMOS FULL-ADDER (Using HALF-ADDER & OR-GATE)
Date Of Performance	:	

BLOCK DIAGRAM , TRUTH – TABLE , BOOLEAN EXPRESSION OF FULL – ADDER

1) Block diagram



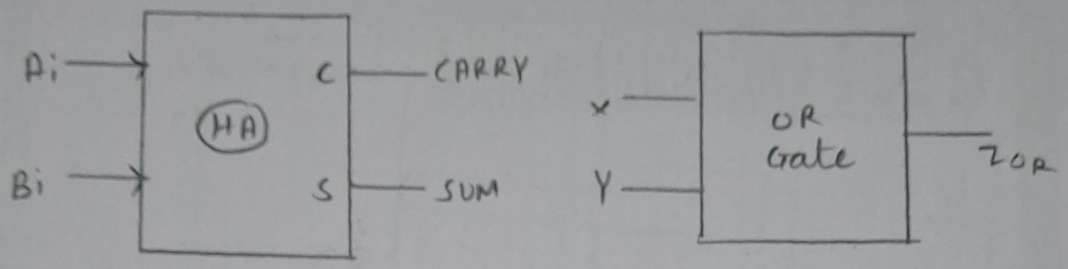
2) Truth table

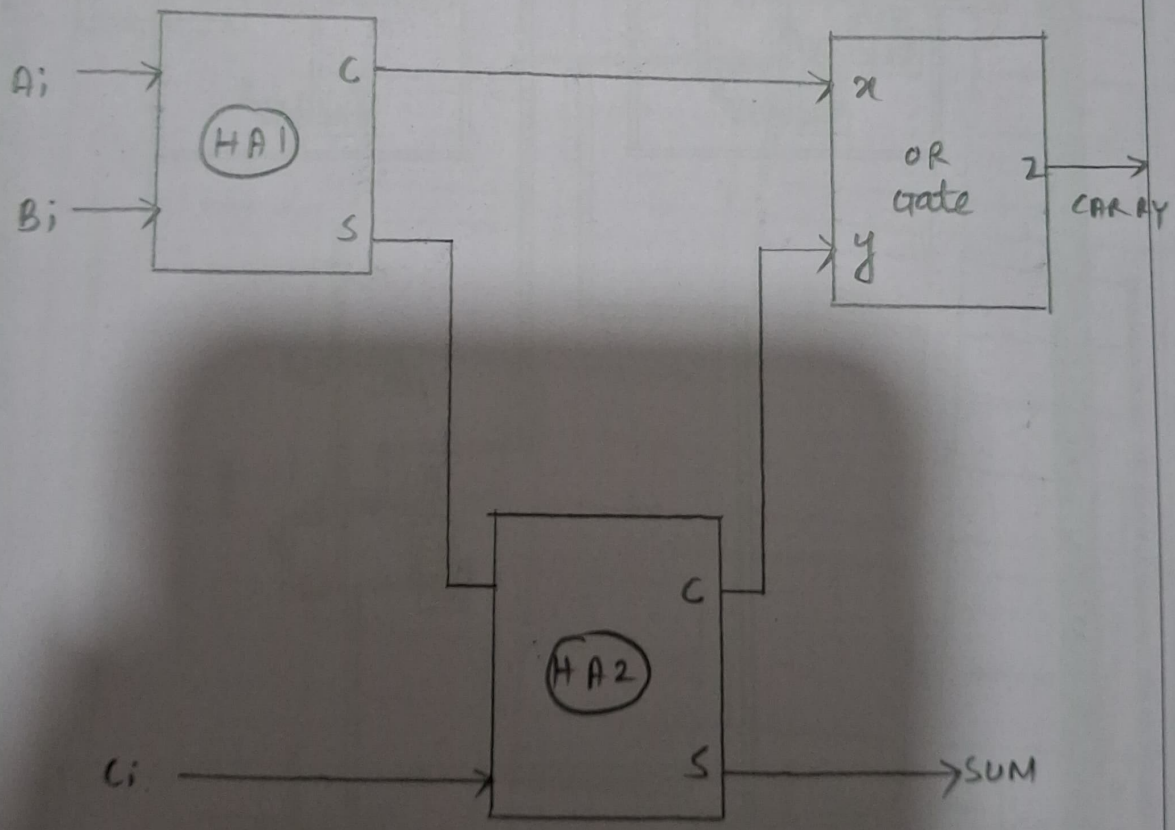
Ai	Bi	Ci	SUM	CARRY
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$SUM = Ai \oplus Bi \oplus Ci$

$CARRY = Ai \cdot Bi + Bi \cdot Ci + Ci \cdot Ai$

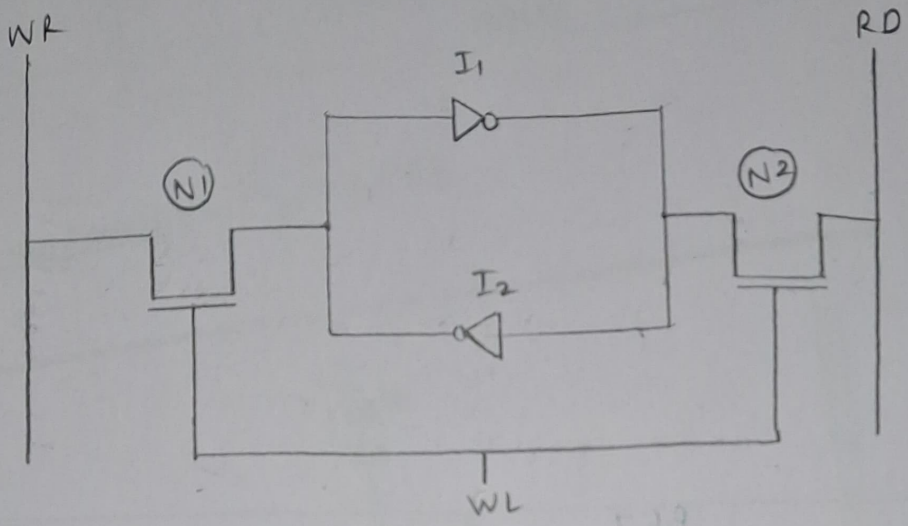
FULL – ADDER USING HALF-ADDER & OR-GATE AS COMPONENTS



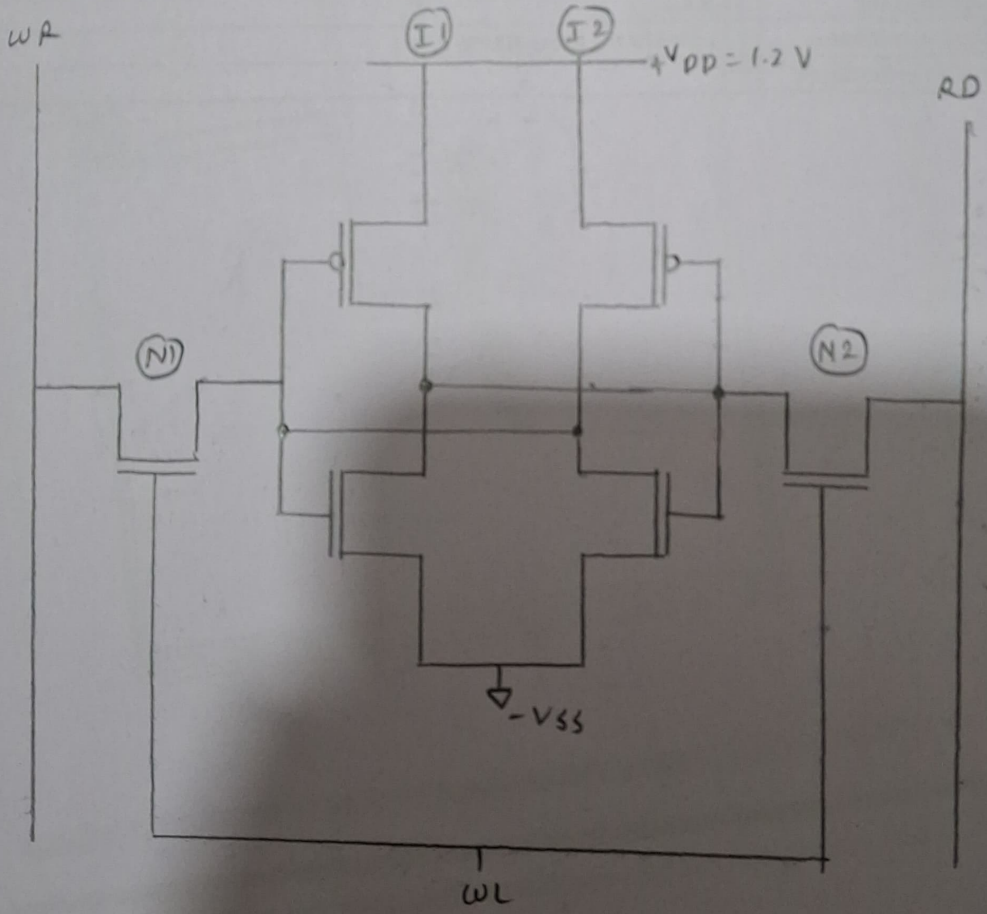


Class	:	BE7
Batch	:	S7
Roll. No	:	42365
ABC ID	:	553725367025
Assignment No.	:	B.4.a
Assignment Name	:	1-bit SRAM (Using NMOS S/W)
Date Of Performance	:	

SYMBOL-LEVEL SCHEMATIC

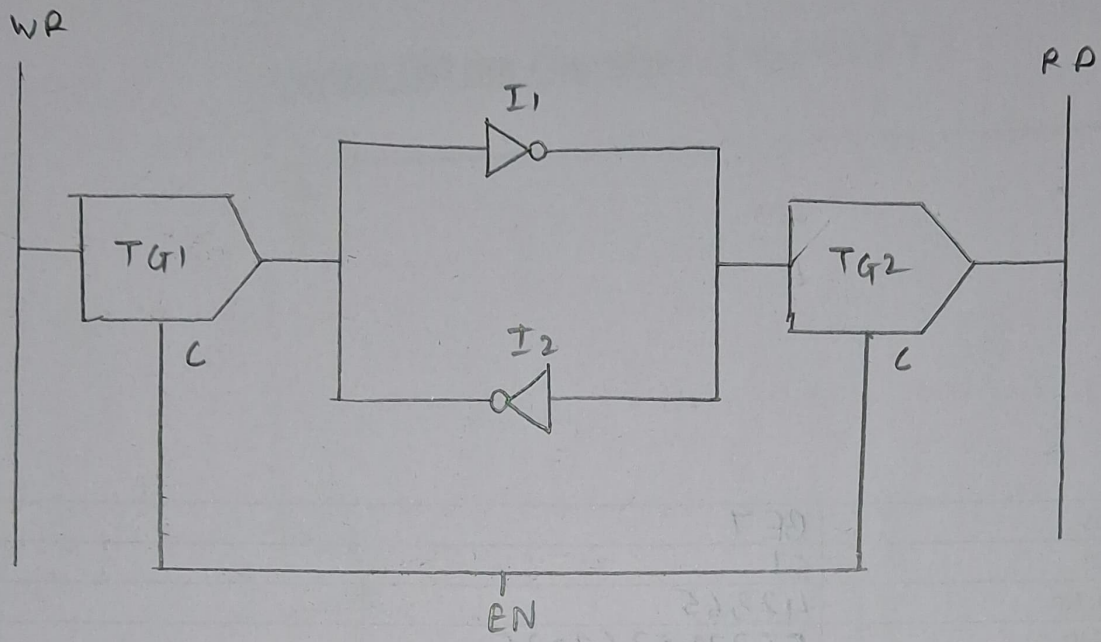


MOSFET-LEVEL SCHEMATIC of 1-bit SRAM Cell Using NMOS S/W :-



Class	:	BE 7
Batch	:	S7
Roll. No	:	42365
ABC ID	:	553725367025
Assignment No.	:	B.4.b
Assignment Name	:	1-bit SRAM (Using TG S/W)
Date Of Performance	:	

SYMBOL-LEVEL SCHEMATIC



MOSFET-LEVEL SCHEMATIC of 1-bit SRAM Cell Using TG S/W :-

